Suggested Teaching Guidelines for

Data Analytics
PGCP-AI February 2026

Duration: 50 hours Classroom + **50** hours Lab + **20** hours Self-Learning

Objective: To introduce the student to Data Analytics

Prerequisites: Knowledge of Programming language of Python Programming and basics of statistics.

Evaluation method: Theory exam– 40% weightage
Lab exam – 40% weightage
Internal exam– 20% weightage

List of Books / Other training material

Courseware: Business Analytics James R Evans, 2nd Edition, Pearson

Reference Book:

1. Business Analytics: Data Analysis &Decision-Making Hardcover – Widescreen
2. Data Science for Business, Foster Provost & Tom Fawcett
3. Data Science and Analytics Paperback by V. K. Jain (Author)
4. Data Analytics for Beginners: A Beginner's Guide to Learn and Master Data Analytics, by Robert J. Woz
5. Data Analytics: Become A Master in Data Analytics by Richard Dorsey

Session 1 & 2: (4T + 4L + 2SL)
Theory:

- Introduction to Business Analytics using some case studies
- Data analytics Life Cycle:
- Discovery
- Data preparation
- Model planning
- Model building implementation
- Quality assurance
- Documentation
- Management approval
- Installation
- Acceptance and operation

Lab Assignment: Setup workspace in either in Google Colab or Jupyter and go through the documentation of Pandas, Numpy, Sklearn libraries.

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Self-Learning: (2SL)

- Read 2–3 business analytics case studies related to retail, finance, healthcare
- Map each case study to Discovery, Data preparation, Model planning, Model building, QA and Deployment & acceptance
- Understand why analytics projects fail (poor data, unclear goals)

Session 3: (2T + 2L + 2SL)

Self-Learning: (2SL)

- Study the use cases of Types of data:
    - Structured, semi-structured, unstructured
    - Numerical vs categorical
- Study Difference between:
    - Reporting
    - Analysis
    - Analytics
- Think what will be the solution When dashboards fail to answer business questions

Theory:

- Intelligent data analysis oNature of Data
- Analytic Processes and Tools Analysis vs. Reporting
- Modern Data Analytic Tools

Lab Assignment: Explore the nature of data.

Session 4: (2T + 2L + 2SL)

Self-Learning: (2SL)

- Explore more about:
    - Misleading visualizations
    - Poor vs effective charts
- Learn which chart to use when
- Understand data storytelling

Theory:

- Visualization and Exploring Data
- Case studies: Making Right Business Decisions based on data

Lab Assignment: Load a sample dataset and explore the data and draw some insights. Also use Matplotlib and seaborn libraries.

Session 5: (2T + 2L + 2SL)
Theory:

- Descriptive Statistical Measures
- Summary Statistics - Central Tendency & Dispersion (Mean, Median, Mode, Quartiles, Percentiles, Range, Interquartile Range, Standard Deviation,
- Variance, and Coefficient of Variation)

Lab Assignment: Load any dataset and find out the mean, median mode and other central tendencies of the dataset.

Self-Learning: (2SL)

- Explore:
    - When mean lies
    - Why median is better in skewed data
- Study Real-world meaning of:
    - Variance
    - Standard deviation
    - Coefficient of variation

Session 6 & 7: (4T + 4L + 2SL)
Theory:

- Sample& population, Uni-variate and bi-variate sampling, re-sampling
- Sample Spaces and Events
- Joint, Conditional and Marginal Probability
- Bayes' Theorem

Lab Assignment: Load any dataset and apply Naive Bayes function and predict the output.

Self-Learning: (2SL)

- Explore the Difference between:
    - Population vs sample
    - Joint vs conditional probability
- Research about Real-life Bayes examples:
    - Medical diagnosis
    - Spam filtering
    - Fraud detection

Session 8: (2T + 2L + 2SL)
Theory:

- Random Variable
- Probability Distribution and Data
- Continuous and discrete distribution – (Normal, Binomial, and Poisson distribution)
- Central Limit Theorem

Lab Assignment: Generate random numbers and check if they are in normal distribution using scipy libraries.

Self-Learning: (2SL)

- Study why normal distribution appears everywhere
- Research and understand When binomial vs Poisson applies
- Explore and study Meaning of Central Limit Theorem (CLT)

Session 9: (2T + 2L)
Theory:

- Sampling and Estimation
- Statistical Interfaces

Lab Assignment: Load any dataset and explore sampling techniques

Session 10: (2T + 2L)
Theory:

- Concepts of Correlation
- Covariance
- Pearson Correlation
- Outliers

Lab Assignment: Load any dataset and find out the covariance between two fields and also find the correlation and determine how two fields are correlated. Also handle the outliers in the data.

Self-Learning: (2SL) (Note: Includes session 9 & 10)

- Sampling bias and its consequences
- Correlation vs causation
- Identify spurious correlations
- Decide wheather to remove, cap & investigate outliers to prevent false conclusions

Session 11 & 12: (4T + 4L +2SL)

- Hypothesis Testing
- Z-Test
- Chi-Square test
- Skewness

Self-Learning: (2SL) (Note: Includes session 11 & 12)

- Explore Logic behind hypothesis testing and Study how it supports data-backed decisions

- Research and Study Meaning of:
    - Null hypothesis
    - p-value
    - Significance
- Explore When to use Z-test vs Chi-square

Session 13: (2T + 2L)

- Predictive modelling and analysis
- Application
- Types
- Benefits and challenges
- The Future of predictive modelling
- The Limitations of Predictive modelling
- Predictive modelling Tools

Session 14, 15 & 16: (6T + 6L)

- Predictive Modelling (From Correlation to Supervised Segmentation):
- Identifying Informative Attributes
- Segmenting Data by Progressive Attributive
- Models Induction and Prediction
- Supervised Segmentation
- Visualizing Segmentations
- Trees as Set of Rules
- Probability Estimation
- Prescriptive Modelling
- Difference between predictive and prescriptive modelling
- How prescriptive analytics works?
- Examples and use cases

Session 17 & 18: (4T + 4L)

- Regression Analysis
- Forecasting Techniques

Self-Learning: (2SL) (Note: Includes session 13 to 18)

- Study how will Predictive vs descriptive vs prescriptive analytics helps in increasing accuracy
- Explore Model assumptions and limitations and understand when will be the results are misleading and when will results are accurate
- Study Regression interpretation (coefficients meaning) in perspective of impact (direction, size etc)

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Session 19 & 20: (4T + 4L)

- Simulation and Risk Analysis
- Optimization- Linear, Nonlinear

Session 21: (2T + 2L)

- Overfitting and Its Avoidance:
- Generalization
- Holdout Evaluation Vs. Cross Validation

Session 22: (2T + 2L)

- Decision Analytics
- Evaluating Classifiers
- Analytical Framework
- Evaluation
- Baseline,
- Performance and Implications for Investments in Data

Session 23: (2T + 2L)

- Evidence and Probabilities
- Explicit Evidence Combination with Bayes Rule
- Probabilistic Reasoning

Session 24: (2T + 2L)

- Factor Analysis
- Directional Data Analytics
- Functional Data Analysis

Session 25: (2T + 2L)

- Introduction to KNIME

Self-Learning: (2SL) (Note: Includes session 19 to 25)

- Study how risk analysis using simulation helps in understanding uncertainty, range of outcomes, and probability of losses or gains, instead of relying on single-point estimates
- Explore optimization intuition by identifying objectives and constraints, and understand how optimal decisions are made when resources, budget, or capacity are limited
- Study overfitting and generalization to understand model reliability, and analyze when a model performs well only on historical data versus when it performs well on unseen data

- Explore model evaluation metrics to assess model performance, and understand which metrics are appropriate for different decision scenarios and why accuracy alone may be misleading
- Study KNIME as a low-code analytics platform to understand analytical workflows, focusing on data flow, decision logic, and outcome interpretation rather than programming details